THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL UNIT TEST SET 1

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. If $\sin\theta = \frac{4}{5}$ then $\sin^2\theta + \cos^2\theta + \tan^2\theta =$ 1

(a) $\frac{16}{9}$ **(b)** $\frac{25}{9}$ **(c)** $\frac{16}{25}$ **(d)** $\frac{9}{25}$

2. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 4 m towards the foot of the tower along a straight line, the angle of elevation becomes 45°. The height of the tower is: 1

(a) $\frac{4\sqrt{3}}{3}$ m **(b)** 4 m **(c)** $6 + 2\sqrt{3}$ m **(d)** $2 + 2\sqrt{3}$ m

3. Tangents PA and PB are drawn from a point P, 50 cm from the centre O of a circle of radius 30 cm. The length of each tangent is 40 cm. The length of the chord AB is: 1

(a) 24 cm **(b)** 48 cm **(c)** 60 cm **(d)** 80 cm

4. A chord of a circle of radius 7 cm subtends an angle of 60° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) The area of the minor segment is: 1

(a) 4.47 cm² **(b)** 21.19 cm² **(c)** 25.67 cm² **(d)** 46.86 cm²

5. If $\tan\theta = \frac{4}{3}$, then $\dfrac{1 - \tan^2\theta}{1 + \tan^2\theta} =$ 1

(a) $-\frac{7}{25}$ **(b)** $\frac{24}{25}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{7}{25}$

6. A flagstaff stands on the top of a building 10 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 30° and 60° respectively. The length of the flagstaff is: 1

(a) 40 m **(b)** $20\sqrt{3}$ m **(c)** 20 m **(d)** 30 m

7. PA and PB are tangents from P to a circle, each of length 7 cm. A third tangent touches the circle at E on the minor arc AB and meets PA at C and PB at D. The perimeter of $\triangle PCD$ is: 1

(a) 9 cm **(b)** 7 cm **(c)** 14 cm **(d)** 21 cm

8. A brooch is a circular piece of silver wire of diameter 21 mm. The wire is also used for 4 equal diameters that divide the brooch into 8 sectors. The total length of wire required (in mm) is: 1

(a) 66 mm **(b)** 108 mm **(c)** 346.5 mm **(d)** 150 mm

9. If $\sin\theta = \frac{4}{5}$, the value of $2\sin\theta\cos\theta$ is: 1

(a) $\frac{12}{25}$ **(b)** $\frac{24}{25}$ **(c)** $\frac{6}{5}$ **(d)** $\frac{8}{5}$

10. A car moves at a uniform speed towards the foot of a vertical tower. The angle of elevation of the top of the tower changes from 30° to 60° in 4 minutes. The further time the car takes to reach the foot of the tower is: 1

(a) $\frac{4}{3}$ minutes **(b)** 8 minutes **(c)** 2 minutes **(d)** 4 minutes

11. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (15x + 4)$ cm and $QB = (8x + 25)$ cm. The length of QA is: 1

(a) 70 cm **(b)** 49 cm **(c)** 3 cm **(d)** 51 cm

12. A horse is tied at the midpoint of one side of a large square field with a rope 35 m long. The area of the field it can graze is: 1

(a) 1925 m² **(b)** 55 m² **(c)** 110 m² **(d)** 3850 m²

13. **Assertion (A):** $\sin^2 60^\circ + \cos^2 60^\circ = 1$. 1
Reason (R): For every angle θ, $\sin^2\theta + \cos^2\theta = 1$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** When an observer walks towards the foot of a tower, the angle of elevation of its top increases. 1
Reason (R): For a tower of fixed height h at horizontal distance d, $\tan\theta = \frac{h}{d}$, which increases as d decreases.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** A tangent PT touches a circle with centre O at P. If $OP = 5$ cm and $PT = 12$ cm, then $OT = 13$ cm. 1
Reason (R): The tangent at any point of a circle is perpendicular to the radius through the point of contact.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. If $\sin\theta = \frac{4}{5}$, find the value of $\cos^2\theta - \sin^2\theta$. 2

17. The angle of elevation of the top of a vertical tower from a point on level ground is 45°. After walking 6 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. Find the height of the tower. 2

18. Tangents PA and PB are drawn from a point P, 75 cm from the centre O of a circle of radius 60 cm. The length of each tangent is 45 cm. Find the length of the chord AB. 2

19. An umbrella has 14 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 49 cm. Find the area between two consecutive ribs. 2

20. If $\cot\theta = \frac{20}{21}$, find $\sin\theta$ and $\sec\theta$. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Prove that: $\tan\theta + \cot\theta = \sec\theta\,\csc\theta$. 3

22. A building is $18\sqrt{3}$ m high. From the top of the building the angle of elevation of the top of a tower is 30° and from the foot of the building it is 60°. Find the height of the tower and its distance from the building (in this order). 3

23. Tangents PA and PB are drawn from a point P, 50 cm from the centre O of a circle of radius 30 cm. The length of each tangent is 40 cm. Find the length of the chord AB. 3

24. A sector of a circle of radius 42 cm has central angle 50°. Find the area of the sector. 3

OR

If $\sin\theta + \cos\theta = \frac{7}{5}$, find $\sin\theta\cos\theta$. 3

25. A flagstaff stands on the top of a building 14 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 45° and 60° respectively. Find the length of the flagstaff. 3

OR

In $\triangle ABC$, right-angled at B, $AB = 36$ cm and $BC = 77$ cm. A circle is inscribed in the triangle. Find the radius of the inscribed circle. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** In a right-angled triangle PQR, the right angle is at Q, with $PQ = 5$ cm, $QR = 12$ cm and $PR = 13$ cm. 5

(i) Write $\sin P$.

(ii) Write $\cos R$.

(iii) Write $\tan P$.

(iv) Verify that $1 + \tan^2 P = \sec^2 P$ for this triangle.

27. Prove that if PA and PB are tangents from P and a third tangent touches the circle at C on the minor arc AB, meeting PA at X and PB at Y, then $XY = AX + BY$ and the perimeter of $\triangle PXY$ is $2PA$. 5

OR

A chord of a circle of radius 35 cm subtends an angle of 60° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) Find the area of the minor segment. 5

END OF QUESTION PAPER — **Total: 50 Marks**